

Dibyakanti Kumar

(+44) 7880 245 191
✉ dibyakanti.kumar@postgrad.manchester.ac.uk
📄 <https://dibyakanti.github.io>
Github: Dibyakanti

Education

Jul 2024 – **The University of Manchester, UK**

Present *Ph.D in Computer Science*

Advisor : Prof. Anirbit Mukherjee, Prof. Alex Frangi

Jul 2018 – **Indian Institute of Technology, Guwahati**

Jul 2022 *Bachelor of Technology in Electronics and Electrical Engineering*

Minor in Computer Science

CGPA – 8.31/10

CGPA – 8.80/10

Experience

Dec 2022– **Research Intern, UNIVERSITY OF MANCHESTER**

July 2024 *Mentored by Prof. Anirbit Mukherjee*

- Assessing the efficacy of Neural Networks in addressing Partial Differential Equations with finite-time blowups.
- Conducting rigorous **theoretical analysis** on existing frameworks to identify potential points of failure.

Aug 2022– **Software Developer, BARCLAYS**

Jun 2024 ○ **Market Risk** | C++ Developer | Apr'23 – Jun'24

- Responsible for maintaining the framework used to compute value-at-risk for various market indices.
- Enhancing **cache efficiency** through the elimination of redundancy in bulk requests and the implementation of multi-threading.
- Improving Solace queue efficiency through the **reduction of message redundancy**.
- **Logging and Monitoring** | Python Developer | Aug'22 – Mar'23
 - Improving the architecture for logging and monitoring for all types of logs
 - Utilizing **ML** to detect **anomaly in logs** and trigger alerts for other teams, to reduce the chance of major interruption of service.

Jun 2020– **Research Intern, UNIVERSITY OF UTAH**

Aug 2022 *Mentored by Dr. Vivek Gupta*

- Semi-automatic rule-based extension of the semi-structured inference dataset **InfoTabS**.
- Introduce **intra-domain counterfactual tables** to discourage **BERT-class models** from learning spurious correlations and recalling pre-train knowledge.
- Incorporated **domain specific constraint** for table validity.
- Improved performance on **InfoTabS** using this dataset as an augmented data.

Jun 2021– **Software Developer Intern, BARCLAYS**

July 2021

- Ever-greening of legacy data ingestion framework and finding viable options for obsolete libraries.
- Utilize multiprocessing libraries like **dask** and **multiprocessing** in python to parallelize data-processing.
- Improved the current data ingestion framework to make them 6 times faster.

Publications

- [1] Generalization Bounds for Physics-Informed Neural Networks for the Incompressible Navier-Stokes Equations
S. Andre-Sloan, **D. Kumar**, A. F Frangi and A. Mukherjee [Paper]
- [2] Convergent Stochastic Training of Attention and Understanding LoRA
Z. Sun, **D. Kumar**, A. F Frangi, A. Mukherjee and M. Sun [Paper]
- [3] Langevin Monte-Carlo Provably Learns Depth Two Neural Nets at Any Size and Data
D. Kumar, S. Jha and A. Mukherjee [Paper]
- [4] Towards Size-Independent Generalization Bounds for Deep Operator Nets
P. Gopalani, S. Karmakar, **D. Kumar** and A. Mukherjee
Published at **TMLR 2024** [Paper]

- [5] Investigating the Ability of PINNs To Solve Burgers' PDE Near Finite-Time BlowUp
D. Kumar and A. Mukherjee
 Published at **IOP-MLST journal** and short version at **NeurIPS 2023** ML4PS workshop [Paper]
- [6] Realistic Data Augmentation Framework for Enhancing Tabular Reasoning. **EMNLP 2022** in Findings.
D. Kumar, V. Gupta, S. Sharma and S. Zhang
 Findings of **EMNLP 2022** [Paper] [Website]

Skills

Languages Python, C++, Julia
 Frameworks JAX, PyTorch, Tensorflow
 Utilities Docker, Git

Relevant Courses

Mathematics Linear Algebra, Multi-variable Calculus, Probability and Random processes, Graphs and Matrices
 ML Pattern Recognition and ML, Natural Language Processing, Data-Driven System Theory
 CS & Others Game Theory, Discrete Mathematics, Data Structure and Algorithm, Operating System, Information Theory, Network Coding and Application, Error Correcting Codes

Services

Organizer **DRSciML 2025 Workshop**
 Co-organized the DRSciML 2025 workshop
 Reviewer AISTATS, ICML, ICLR, Neurocomputing Journal, IOP-MLST Journal
 Opensource **IITG.AI 2020–21**
 Head AI and ML community at IIT Guwahati

Achievements

2020 Recipient of **Silver Medal** at Inter-IIT TechMeet DataScience Competition
 2018 Ranked among the **top 2%** in JEE Advanced held for 0.15 million candidates